

Quantum Key Distribution (QKD) System Utilizing Super- positional Entropy for Post- Quantum Cryptography

Inventor: Jed Kircher

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Abstract

Quantum Key Distribution (QKD) System Utilizing Superpositional Entropy for Post-Quantum Cryptography

This invention introduces a Quantum Key Distribution (QKD) system that utilizes superpositional entropy—a novel, intrinsic source of quantum randomness—to generate, validate, and securely exchange cryptographic keys. Unlike conventional QKD systems that rely on physical quantum random number generators (QRNGs), which may suffer from hardware bias and environmental noise, this approach extracts entropy directly from the probabilistic structure of quantum superposition states, ensuring unbreakable encryption security even in the presence of quantum computing adversaries.

The system consists of three core components:

1. **Superpositional Entropy Generator (SEG):** Extracts entropy from quantum wavefunction probability distributions, ensuring cryptographic keys remain unpredictable and information-theoretically secure.
2. **Secure Quantum Transmission & Entropy Validation:** Uses quantum-state-based key exchange with embedded entropy markers to detect tampering or eavesdropping attempts.
3. **Post-Quantum Secure Key Exchange Protocol:** Introduces self-correcting entropy validation, eliminating the need for classical post-processing, which is often a weak point in traditional QKD systems.

By integrating superpositional entropy into quantum cryptographic key exchange, this system ensures:

- Resilience against quantum decryption attacks, including Shor's Algorithm and Grover's Algorithm.
- Elimination of side-channel vulnerabilities through tamper-sensitive entropy markers.
- Scalability for post-quantum applications, including blockchain security, government communications, and global satellite QKD networks.

This innovation represents a paradigm shift in quantum cryptography, setting a new standard for long-term, quantum-safe encryption by leveraging the fundamental unpredictability of quantum superposition rather than relying on imperfect physical QRNGs.

1. Background of the Invention

1.1 The Need for Secure Key Distribution

Modern encryption systems rely on secure key exchange protocols to protect sensitive data. Traditionally, this is achieved using public-key cryptography (e.g., RSA, ECC, Diffie-Hellman). However, these methods face an existential threat from quantum computing, as algorithms like Shor's Algorithm can efficiently break classical encryption.

1.2 Quantum Key Distribution (QKD) as a Solution

QKD offers a theoretically unbreakable encryption method by leveraging quantum mechanics principles to exchange cryptographic keys. Existing QKD protocols, such as BB84 and E91, rely on quantum states (e.g., photon polarization) to detect eavesdropping and guarantee secure transmission.

However, current QKD implementations have three critical limitations:

1. Imperfections in Quantum Random Number Generators (QRNGs)
 - Many QKD systems rely on QRNGs that are influenced by external noise, hardware biases, and environmental factors. This introduces statistical predictability, reducing security.
2. Vulnerabilities to Quantum Adversaries
 - While QKD ensures key secrecy under ideal conditions, some implementations remain vulnerable to quantum side-channel attacks, where attackers exploit biases in key randomness.
3. Post-Quantum Cryptographic Challenges
 - Even though QKD is quantum-safe, it still interacts with classical error correction and authentication protocols that may be vulnerable to quantum attacks.

1.3 Introduction of Superpositional Entropy for Enhanced QKD Security

This invention overcomes these challenges by introducing Superpositional Entropy, a method that extracts entropy directly from quantum superposition fluctuations, ensuring that cryptographic keys:

- Remain fundamentally unpredictable, even under quantum adversaries.
- Are resistant to side-channel attacks, since their randomness source is entangled with quantum probability itself.
- Improve error correction and authentication security, as the keys dynamically self-adjust based on quantum entropy validation.

This superpositional entropy approach fundamentally strengthens QKD implementations, making them practical for post-quantum encryption at scale.

2. Summary of the Invention

The present invention introduces a Quantum Key Distribution (QKD) system that utilizes superpositional entropy to enhance cryptographic key security, ensuring resilience against quantum computing attacks.

Unlike conventional QKD implementations that rely on finite entropy sources from physical QRNGs, this invention directly extracts entropy from the probabilistic nature of quantum superposition states. By leveraging the inherent unpredictability of superpositional entropy, the invention achieves the following breakthroughs in quantum cryptography:

2.1 Core Innovations

1. Superpositional Entropy for Quantum Key Randomness
 - This method extracts entropy from the underlying probability distribution of quantum superposition states, ensuring that generated encryption keys remain fundamentally unpredictable.
 - Unlike conventional randomness sources (e.g., vacuum fluctuations, phase noise in photonic QRNGs), superpositional entropy is an intrinsic quantum property that is mathematically unforgeable.

2. Dynamic Self-Adjusting Entropy Correction
 - The system employs a self-correcting entropy validation process, which detects and eliminates biases in real-time, strengthening randomness quality.
 - This method ensures that cryptographic keys are not only generated with quantum security but also validated to meet strict post-quantum cryptographic standards.
3. Post-Quantum Secure Key Exchange Protocols
 - By embedding superpositional entropy markers in each quantum-generated key, the system can authenticate randomness integrity at both the sender and receiver ends.
 - This eliminates weaknesses in classical post-processing steps (such as error correction and authentication), ensuring that QKD security extends beyond transmission into cryptographic protocol security.

2.2 Functional Benefits of the Invention

1. Unbreakable Cryptographic Key Generation
 - The unpredictability of superpositional entropy ensures that encryption keys cannot be predicted or reconstructed, even with quantum computational power.
2. Resistance to Side-Channel and Entropy Harvesting Attacks
 - Since the entropy source exists within the fundamental nature of quantum superposition, it is immune to classical and quantum side-channel attacks, where adversaries attempt to extract biases from physical randomness generators.
3. Improved Error Correction & Authentication
 - The system integrates an entropy-based error correction mechanism, where deviations in quantum entropy values serve as an authentication measure to detect unauthorized tampering or eavesdropping.
4. Scalable for Global Quantum Networks & Cryptographic Applications

- The invention is designed for practical deployment in global-scale QKD networks, including applications in financial security, military communications, and blockchain-based post-quantum security.

2.3 Summary of Patent Claims

This invention claims:

1. A Quantum Key Distribution (QKD) system that utilizes superpositional entropy to enhance encryption key randomness, ensuring post-quantum security.
2. A method for extracting higher-dimensional entropy from quantum superposition fluctuations to refine randomness in QKD key generation.
3. An adaptive QKD protocol that self-corrects entropy deviations using superpositional entropy markers, preventing side-channel attacks.
4. A post-quantum cryptographic system where QKD keys incorporate entropy markers from quantum superposition, preventing algorithmic decryption via quantum computing techniques.

3. Detailed Description of the Invention

The present invention introduces a Quantum Key Distribution (QKD) system utilizing superpositional entropy, a novel approach to generating, validating, and distributing cryptographic keys with post-quantum security. This section details:

- The mechanics of superpositional entropy extraction.
- How it enhances QKD security against quantum and classical attacks.
- The system architecture and real-world implementation.

3.1 Superpositional Entropy: A Novel Source of Cryptographic Randomness

Superpositional entropy is derived from the fundamental probabilistic nature of quantum superposition states. Unlike traditional quantum random number generators (QRNGs) that extract entropy from physical measurements of vacuum fluctuations, thermal noise, or

spontaneous emission, this method directly harnesses the intrinsic stochastic behavior of quantum wavefunctions in superposition.

Mathematical Formulation of Superpositional Entropy

Let $|\psi\rangle$ be a quantum state in superposition:

$$|\psi\rangle = \sum_i c_i |i\rangle$$

where:

- $|i\rangle$ are basis states, and
- c_i are probability amplitudes that satisfy $\sum_i |c_i|^2 = 1$.

The entropy associated with this state is given by:

$$\mathcal{H}(|\psi\rangle) = - \sum_{i=1}^n |c_i|^2 \log |c_i|^2$$

This entropy function is inherently probabilistic, unlike classical entropy sources that may be biased by external factors. It provides a pure quantum entropy source that remains unpredictable, even in the presence of quantum computational adversaries.

Unlike conventional QRNGs that rely on physical randomness extraction methods, superpositional entropy is an intrinsic feature of quantum mechanics, making it immune to hardware defects, environmental noise, and quantum side-channel attacks.

3.2 Superpositional Entropy vs. Traditional QRNG-Based QKD

Existing Quantum Key Distribution (QKD) systems primarily rely on Quantum Random Number Generators (QRNGs) to generate encryption keys. These QRNGs extract randomness from physical quantum processes, such as:

- Vacuum fluctuations (measuring random photon phase shifts)
- Thermal noise (Johnson-Nyquist noise in superconducting circuits)
- Spontaneous photon emission

However, physical QRNGs introduce inherent weaknesses that compromise QKD security, particularly in a post-quantum threat environment.

Limitations of QRNG-Based QKD Systems

1. Predictability from Hardware Biases
 - QRNGs rely on physical measurements, meaning they can introduce statistical biases due to imperfections in quantum sensors, optical components, and external environmental conditions.
 - Adversaries can exploit these biases to predict QRNG output with high confidence.
2. Vulnerabilities to Side-Channel Attacks
 - QRNG-based QKD systems are vulnerable to entropy leakage, where adversaries monitor system behavior to infer key randomness patterns.
 - Even with cryptographic post-processing, these subtle leaks over time can lead to security vulnerabilities.
3. Limited Scalability for Post-Quantum Security
 - QRNG randomness quality is dependent on hardware fidelity, meaning long-distance QKD networks (e.g., satellite-to-ground or global financial networks) risk entropy degradation.

Superpositional Entropy: A Superior Alternative

This invention completely bypasses QRNG weaknesses by introducing superpositional entropy, which derives randomness from the probabilistic nature of quantum superposition itself.

Key Innovations in Superpositional Entropy-Based QKD

- Entropy is intrinsic to quantum wavefunction behavior, eliminating dependency on external physical sensors.
- Quantum states collapse unpredictably, meaning adversaries cannot extract bias patterns or infer key randomness.
- Self-correcting entropy validation prevents entropy leakage, eliminating traditional side-channel vulnerabilities.

Mathematical Justification of Superpositional Entropy

Let a quantum state $|\psi\rangle$ exist in superposition:

$$|\psi\rangle = \sum_i c_i |i\rangle$$

where:

- $|i\rangle$ are basis states, and
- c_i are probability amplitudes such that $\sum_i |c_i|^2 = 1$.

The entropy associated with the superposition is given by:

$$\mathcal{H}(|\psi\rangle) = - \sum_{i=1}^n |c_i|^2 \log |c_i|^2$$

Unlike physical QRNG entropy sources, which are prone to external biases, superpositional entropy remains inherently unpredictable because:

- It is intrinsic to quantum mechanics, rather than derived from classical measurements.
- The system does not interact with external noise sources, ensuring absolute entropy purity.
- Collapsing the wavefunction instantly destroys any prior entropy predictions, ensuring maximum randomness protection.

Comparison of Randomness Sources in QKD Systems

Table 1

Feature	QRNG-Based QKD	Superpositional Entropy-Based QKD
<u>Entropy Source</u>	Physical QRNGs (vacuum fluctuations, phase noise)	Quantum superposition probability distributions
<u>Predictability</u>	Hardware imperfections may introduce bias	Fundamentally unpredictable by nature
<u>Side-Channel Security</u>	Vulnerable to entropy leakage & quantum side-channel attacks	Immune to classical & quantum side-channel attacks
<u>Post-Quantum Resilience</u>	Limited long-term security against quantum decryption	Intrinsically secure against all known quantum attacks
<u>Implementation Complexity</u>	Requires additional post-processing & error correction	Self-correcting entropy validation—no extra post-processing needed

This invention ensures post-quantum cryptographic resilience by eliminating entropy predictability, making it infeasible for quantum adversaries to reconstruct encryption keys.

Having established the fundamental advantages of superpositional entropy, we now describe its integration into a secure QKD system.

3.3 QKD System Architecture: Hardware & Software Components

The system consists of three primary components:

(1) Superpositional Entropy Generator (SEG)

- Extracts entropy from quantum superposition wavefunctions.
- Implements a probabilistic entropy validation function to ensure maximum unpredictability of generated keys.
- Uses entropy markers that self-collapse if external manipulation occurs.

(2) Quantum Key Distribution & Secure Transmission

- Secure transmission via polarized photon states, entangled qubits, or superconducting quantum circuits.
- Real-time entropy monitoring: Detects deviations in expected entropy values to identify potential quantum side-channel attacks.
- Uses post-quantum cryptographic protocols that eliminate the need for classical post-processing (error correction, authentication).

(3) Quantum Key Validator & Authentication Unit

- Verifies that exchanged quantum keys maintain high entropy and randomness integrity.
- Implements self-correcting entropy markers, ensuring keys remain unpredictable over extended transmissions.
- Detects tampering attempts based on entropy fluctuation analysis.

Illustrative QKD System Workflow

1. Quantum Key Generation

- The SEG extracts entropy from quantum superpositions and generates an initial quantum-secure key (K_{raw}).

2. Entropy Validation & Bias Correction

- The system evaluates entropy fluctuations in real-time:

$$K_{\text{validated}} = K_{\text{raw}} \oplus \mathcal{F}(\mathcal{H}(|\psi\rangle))$$

where $\mathcal{F}(\mathcal{H})$ is an adaptive function that corrects for entropy inconsistencies, ensuring pure quantum randomness in the final key.

3. Quantum Key Exchange via Secure Channel

- The validated key is transmitted using a QKD protocol (e.g., BB84, E91, MDI-QKD).
- Any attempt to extract information from the key disrupts the entropy markers, instantly signaling an eavesdropping attempt.

4. Quantum Key Authentication & Use in Cryptographic Applications

- The receiver re-validates the entropy markers to confirm key integrity.
- Secure keys are then used in quantum-safe encryption protocols for high-security communications.

3.4 Self-Correcting Entropy Validation: Attack Detection & Prevention

A key innovation in this QKD system is the self-correcting entropy validation mechanism, which prevents both classical and quantum entropy-harvesting attacks by embedding entropy markers within cryptographic keys. Unlike traditional QKD post-processing methods that rely on classical error correction—which itself can be vulnerable to quantum computational attacks—this approach ensures that only keys with fully validated superpositional entropy are accepted, making them unpredictable and resistant to entropy manipulation.

Mathematical Model of Entropy Validation

Let K_{raw} be the initially generated quantum-secure key extracted from superpositional entropy, prior to validation:

$$K_{\text{raw}} = \{k_1, k_2, \dots, k_n\}$$

where k_i are individual key bits derived from quantum wavefunction probability distributions.

Each k_i is associated with an entropy marker that validates its integrity. The entropy function of the key is given by:

$$\mathcal{H}(K_{\text{raw}}) = - \sum_{i=1}^n P(k_i) \log P(k_i)$$

where:

- $P(k_i)$ is the probability of measuring key bit k_i in a given quantum basis.
- The function ensures that only keys with maximal quantum entropy pass validation.

The self-correction function is then applied:

$$K_{\text{validated}} = K_{\text{raw}} \oplus \mathcal{F}(\mathcal{H}(K_{\text{raw}}))$$

where:

- $\mathcal{F}(\mathcal{H})$ is an adaptive entropy correction function that detects entropy deviations and applies necessary corrections.
- Bitwise XOR operation ($\oplus$) prevents direct extraction of entropy markers by potential adversaries.

How This Prevents Quantum Side-Channel Attacks

A major concern in QKD security is entropy leakage, where an adversary (Eve) attempts to:

- Predict key randomness by analyzing hardware-induced biases in key generation.

- Inject entropy manipulation attacks to subtly alter the probability distribution of cryptographic key bits.

This invention prevents such attacks by embedding an entropy marker verification step at both the sender and receiver ends.

Attack Detection Process

1. Quantum Key Generation

- Alice generates K_{raw} from superpositional entropy and applies entropy marker embedding.
- The probability distribution of the key bits is recorded, forming a baseline for validation.

2. Quantum Key Exchange via Secure Channel

- The quantum-secure key is transmitted using tamper-sensitive QKD protocols (BB84, E91, MDI-QKD, etc.).
- Any adversary attempting to read or extract entropy information collapses the quantum state, introducing detectable deviations.

3. Entropy Marker Verification & Validation at Receiver End

- Bob verifies K_{received} against the expected entropy function:

$$\mathcal{H}(K_{\text{received}}) \stackrel{?}{=} \mathcal{H}(K_{\text{raw}})$$

- If entropy markers remain intact, the key is accepted for encryption.
- If entropy deviations exceed threshold tolerances, the key is rejected as compromised.

Final Security Guarantees

- Prevents Predictability-Based Attacks: Any attempt to predict entropy markers or key randomness is computationally infeasible.
- Eliminates Side-Channel Attacks: Any adversarial interaction collapses the entropy markers, triggering key rejection.

- Ensures Long-Term Post-Quantum Security: The system does not rely on classical error correction, making it resilient to quantum decryption methods (e.g., Shor's Algorithm, Grover's Algorithm).

3.5 Post-Quantum Security: Resistance to Quantum Attacks

This invention ensures security against all known quantum computational attacks:

1. Unpredictable Key Generation

- Superpositional entropy provides an information-theoretically secure randomness source, preventing any predictive attacks, even with unlimited quantum computational power.

2. Immunity to Quantum Side-Channel Attacks

- Classical QRNG-based QKD systems are vulnerable to entropy leakage, where adversaries exploit imperfections in key randomness.
- This invention eliminates this weakness by embedding entropy markers that collapse upon adversarial interaction.

3. No Classical Post-Processing Weaknesses

- Traditional QKD relies on classical error correction & authentication protocols, which may introduce vulnerabilities.
- The proposed system eliminates these weaknesses by validating entropy markers as a built-in security measure.

3.6 Implementation Pathways & Industry Deployment

This QKD system is designed for scalability and integration into real-world cryptographic infrastructures, including:

- Government & Defense – Securing classified communications from quantum decryption threats.
- Financial Institutions – Protecting high-value transactions with quantum-secure encryption.

- Blockchain & Post-Quantum Cryptography – Enabling quantum-resistant blockchain-based security frameworks.
- Telecommunications – Integrating superpositional entropy into quantum-secure 5G, satellite, and fiber-optic networks.

4. Claims

The following claims define the scope of the invention and ensure broad yet enforceable patent protection. These claims focus on superpositional entropy as a novel cryptographic security mechanism within Quantum Key Distribution (QKD) systems.

4.1 Independent Claims

1. A Quantum Key Distribution (QKD) system comprising:

- a superpositional entropy generator (SEG) configured to extract entropy from quantum superposition probability distributions;
- a secure quantum transmission channel utilizing quantum states for key distribution; and
- a key validation and authentication mechanism that verifies randomness integrity through entropy markers, wherein the system ensures post-quantum security by preventing predictability in cryptographic key generation and distribution.

2. A method for generating quantum-secure cryptographic keys, comprising:

- extracting entropy from quantum superposition states;
- applying an entropy validation function to detect and correct randomness biases; and
- embedding superpositional entropy markers in the generated key, wherein said method ensures resistance to quantum computational attacks and entropy-harvesting adversaries.

3. A post-quantum cryptographic system, comprising:

- a quantum key generator implementing superpositional entropy-based randomness extraction;

- a secure QKD protocol utilizing entropy-validated quantum keys; and
- an adaptive entropy authentication process that dynamically validates quantum randomness, wherein the system achieves quantum-resistant encryption through inherently unpredictable cryptographic keys.

4.2 Dependent Claims

4. The QKD system of claim 1, wherein the superpositional entropy generator (SEG):

- utilizes a probability amplitude entropy function to dynamically adjust key randomness; and
- prevents entropy leakage and quantum side-channel attacks through continuous randomness monitoring.

5. The QKD system of claim 1, wherein the key validation mechanism:

- applies a self-correcting entropy function to enhance key unpredictability, and
- detects any external interference in quantum key exchange by monitoring entropy deviations.

6. The method of claim 2, wherein the entropy extraction process:

- utilizes a probability function $\mathcal{H}(|\psi\rangle) = - \sum_i |c_i|^2 \log |c_i|^2$ to compute entropy from superpositional states, and
- prevents classical and quantum biases in randomness generation.

7. The method of claim 2, wherein embedding superpositional entropy markers in the cryptographic key:

- ensures authenticity by encoding non-reproducible entropy signatures, and
- dynamically adapts to changes in entropy levels during quantum key exchange.

8. The post-quantum cryptographic system of claim 3, wherein the quantum-secure protocol:

- is implemented in a fiber-optic, satellite, or superconducting quantum network, and
- ensures cryptographic key resilience against future quantum decryption attacks.

9. The post-quantum cryptographic system of claim 3, wherein the quantum key distribution mechanism is integrated into a blockchain-based encryption framework, ensuring:

- Quantum-resistant cryptographic ledger security through verifiably random, quantum-secure keys.
- Decentralized post-quantum authentication, preventing unauthorized modifications to digital records.

10. The post-quantum cryptographic system of claim 3, wherein the quantum key distribution process is deployed via a satellite-based network, ensuring:

- Global-scale quantum-secure communication using orbiting quantum key exchange nodes.
- Long-range cryptographic key synchronization, enabling quantum security across geographically distributed locations.

4.3 Strategic Claim Considerations

- **Broad Scope:** Covers any QKD system utilizing superpositional entropy without limiting to specific hardware implementations.
- **Difficult to Design Around:** Ensures that any method using quantum superposition probability distributions for key entropy falls under this patent.
- **Post-Quantum Security Emphasis:** Clearly differentiates from existing QKD patents by demonstrating intrinsic unpredictability of cryptographic keys via superpositional entropy.